



AI IDEA FORGE

— AN AI-POWERED PRODUCT DISCOVERY FRAMEWORK —



RESEARCH



COMPETITORS



GAP ANALYSIS



FEASIBILITY



RECOMMENDATIONS



BATCH 4

| CAPSTONE PROJECT

| GROUP 8



Group 8 Members

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AI Idea Forge

An AI-Powered Product Discovery Framework

Tagline: *Transform Ideas into Validated Products with AI*

Problem Statement

Validating a product idea is often a manual, time-consuming, and inconsistent process. Teams typically use multiple tools to research competitors, analyze market trends, evaluate technical feasibility, and assess business potential. This fragmented approach increases research effort, delays decision-making, and can lead to costly mistakes.

Proposed Solution

AI Idea Forge provides a centralized AI-powered platform that accepts a product idea, business concept, or website URL and automatically performs comprehensive product discovery using specialized AI agents and live web research.

The platform delivers structured insights and actionable recommendations, enabling users to determine whether an idea is worth pursuing before development begins.

Features to Build

- AI Idea Analysis
 - Live Market Research
 - Competitor Discovery
 - Competitor Benchmarking
 - Market Gap Analysis
 - Technical Feasibility
 - Business Feasibility
 - AI Recommendations
 - Executive Report Generation
 - Interactive Dashboard
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Target Users

- Entrepreneurs
- Startup Founders
- Product Managers
- Innovation Teams
- Business Analysts
- Digital Marketer
- Students and Capstone Teams

Technology Stack

LAYER	TECHNOLOGY
FRONTEND	Lovable/Replit
WORKFLOW ORCHESTRATION	n8n
AI MODELS	OpenAI GPT/Claude
DATABASE	Supabase
LIVE RESEARCH	Web Search APIs
BACKEND	REST APIs
DOCUMENTATION	GitHub
API TESTING	Postman/Vercel

Impact

AI Idea Forge reduces the effort required to validate product ideas by automating research and analysis through a repeatable AI workflow. It enables faster, more informed product decisions while demonstrating AI orchestration, workflow automation, API integration, and modern software architecture.